ETH Flow Monitor — Product Requirements Doc

Oct 14, 2026 · @Nowis (Yuchen Zhou) · FinTech-564, Glamsterdam project

Overview

ETH Flow Monitor is a real-time API and dashboard that shows every movement of native ETH on Ethereum, including transfers made inside smart contracts. It also alerts on addresses a user is watching and screens each flow against sanctions lists.

The product depends on one Glamsterdam change. EIP-7708 makes every ETH transfer emit a standard Transfer log, the same way ERC-20 tokens already do. Today, tracking ETH sent from a multisig or smart wallet takes expensive call tracing. After Glamsterdam, a plain log subscription is enough.

The MVP has three parts:

1. **Real-time ETH transfer feed:** every ETH Transfer log, decoded and streamed.
2. **Watched-address alerts:** a notification when a watched address sends or receives ETH.
3. **Basic sanctions screening:** each transfer checked against the OFAC SDN crypto-address list, with any hit flagged.

Problem statement

When ETH moves inside a smart contract, such as a multisig, a smart wallet or a DeFi protocol, the chain records no log of it. Exchanges and compliance teams have only two ways to see that ETH. They can run expensive tracing infrastructure that re-executes every transaction, or they can buy the data from a large analytics vendor. If they do neither, they miss the flow.

This matters for three reasons:

- **Visibility gap:** ERC-20 tokens are easy to track because every transfer emits a log. Native ETH is not.
- **Cost:** call tracing needs archive or trace-enabled nodes, and that cost locks smaller exchanges, custodians and compliance teams out.

- **Compliance risk:** sanctioned or stolen funds can pass through contracts unnoticed, and an exchange that misses them takes on regulatory and reputational risk.

Glamsterdam's EIP-7708 closes the gap at the protocol level. Every ETH transfer becomes a standard, indexable log. Monitoring that used to need specialist infrastructure can now run on ordinary log subscriptions. ETH Flow Monitor turns those new logs into a feed, alerts and screening that a small team can afford.

Goals and non-goals

Goals

1. Show every native ETH transfer, including transfers made inside contracts, using only EIP-7708 logs and no call tracing.
2. Alert users within seconds when a watched address sends or receives ETH.
3. Flag any transfer that touches an address on the OFAC sanctions list.
4. Prove the idea on the Sepolia testnet: a contract-internal ETH transfer that is invisible in today's logs appears as a Transfer event after Glamsterdam.
5. Keep it cheap enough to run on a standard RPC endpoint, with no archive or trace node.

Non-goals

- **Full blockchain forensics:** no address clustering, entity attribution or risk scores like Chainalysis or TRM. The MVP flags; it does not investigate.
- **ERC-20 and NFT tracking:** existing tools already cover these well.
- **Other chains and L2s:** Ethereum mainnet and Sepolia only.
- **History before Glamsterdam:** those blocks have no 7708 logs, so the MVP starts at the fork.
- **Blocking or freezing funds:** the product informs a human. It never acts on funds or files reports.
- **A licensed compliance service:** the MVP supports a compliance team's work but makes no legal determinations.

Target personas

Persona	Role	Pain today	How ETH Flow Monitor helps
Maya, compliance analyst	AML/sanctions analyst at a mid-size exchange	Misses ETH routed through contracts. Pays for a large vendor she can only partly use.	Sanctions hits on every ETH flow, including contract-internal ones, in one alert queue.
Dev, exchange backend engineer	Runs deposit and withdrawal infrastructure	Deposits sent from smart wallets need traces to detect. Trace nodes are slow and costly.	One log-subscription API instead of a trace pipeline.
Priya, custody operations lead	Watches client and treasury wallets at a custodian	Can't see in real time when ETH leaves a client's multisig through a contract call.	Watched-address alerts within seconds, with a link to the transaction.

User stories

- As **Maya**, I want every ETH transfer touching a sanctioned address to be flagged, even when it happens inside a contract, so I don't miss exposure that today's logs hide.
- As **Maya**, I want each alert to show the sender, receiver, amount, transaction hash and matched list entry, so I can decide to escalate in under a minute.
- As **Dev**, I want to subscribe to ETH transfers into our deposit addresses through one API, so I can credit smart-wallet deposits without running a trace node.
- As **Priya**, I want to add a list of client wallets and get an alert within seconds of ETH leaving any of them, so I can confirm the client approved the move.
- As **Priya**, I want to export an address's ETH flow history as CSV, so I can attach it to an audit or incident report.

Requirements

Functional

ID	Requirement	Priority
F1	Ingest every new block and decode all EIP-7708 ETH Transfer logs (from, to, amount, tx hash, block, timestamp). Accept only logs emitted by the system address 0xfff…fffe. ERC-20 Transfer events use the same topic, so this filter keeps token transfers from being counted as ETH.	Must
F2	Mark whether each transfer was top-level (a normal transaction) or contract-internal.	Must
F3	Stream the decoded transfers through a live dashboard feed and a WebSocket API.	Must
F4	Let users add, label and remove watched addresses, one at a time or by CSV upload.	Must
F5	Alert on any transfer into or out of a watched address, with an optional minimum amount.	Must
F6	Screen both sides of every transfer against the ETH addresses on the OFAC SDN list (the "Digital Currency Address - ETH" fields), refreshed daily. OFAC says its address listings are not exhaustive, so a clean result does not prove an address is safe.	Must
F7	Send alerts in the dashboard, by email and by webhook.	Must
F8	Let an analyst mark an alert as reviewed, escalated or a false positive, with a note.	Should
F9	Search and filter transfer history by address, amount and date, and export it as CSV.	Should
F10	Demo mode: show the same Sepolia transaction before and after Glamsterdam, side by side.	Should
F11	REST API to read transfers and alerts.	Could

Non-functional

Area	Requirement
Performance	Alert within 5 seconds of a block being seen. Keep up with peak mainnet load without falling behind, with headroom for higher gas limits. EIP-8261 lets mainnet gas limits rise on a schedule after Glamsterdam, but no values are set yet.
Reliability	Miss no blocks: track the last block processed and backfill gaps after a restart. Handle reorgs by rolling back and re-emitting affected transfers. 99.5% uptime target.
Data accuracy	Each day, ETH transfer totals match an independent node query for a sample of blocks.
Security	API keys per user, all traffic over HTTPS, watchlists visible only to the team that owns them. No private keys are ever handled.
Privacy	Watchlists and alert notes are customer-confidential and encrypted at rest. They are never shared or used across customers.
Compliance	Record the sanctions-list version used for every screening decision, so a result can be audited later. Label the product as decision support, not a legal determination.
Accessibility	The dashboard meets WCAG 2.1 AA: keyboard navigation, alert severity shown with text as well as color, readable contrast.
Cost	Runs on one standard RPC provider plan. No archive or trace node required.

Success metrics

The main test is whether the monitor finds every ETH transfer a trace-based tool finds, at a fraction of the cost. The targets below are MVP targets for the Sepolia prototype, not measured results.

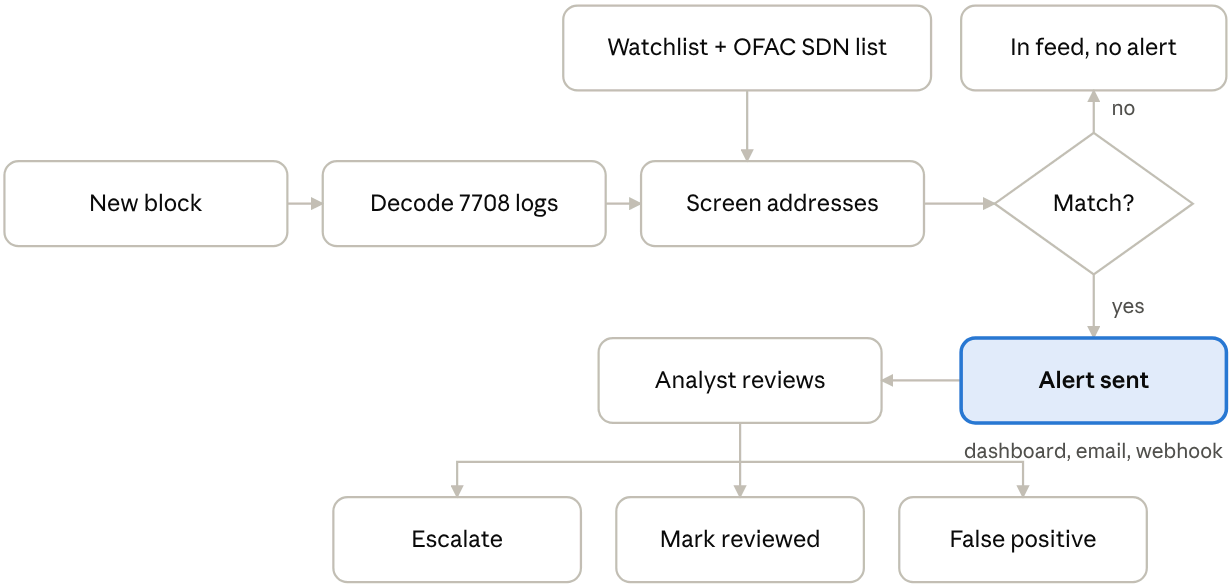
KPI	How it's measured	MVP target
Coverage vs. tracing	Transfers found from 7708 logs ÷ transfers found by `debug_traceBlock` on the same blocks	100%
Contract-internal transfers surfaced	Transfers caught that today's logs miss, in a test set of multisig and smart-wallet sends	100% of the test set
Alert latency	Time from block seen to alert delivered (p95)	under 5 s
Missed blocks	Blocks never processed, per day	0
Sanctions screening recall	Share of replayed test transfers that get flagged. The test set uses the 15 SDN-listed ETH addresses of Lazarus Group (8) and Garantex Europe (7). In the live Sepolia demo, a mock sanctions entry stands in for them, so no ETH is ever sent to a real sanctioned address.	100%
False-positive rate	Alerts an analyst marks as false positives ÷ all alerts	under 5%
Infrastructure cost	Monthly RPC and hosting cost, compared with running a trace node	at least 10x cheaper
Analyst time per alert	Time from alert opened to marked reviewed or escalated, in user testing	under 1 min

After the class project, adoption would be measured by active watchlists, alerts reviewed each week, and pilot users still active after 30 days.

User experience / design

Flow

Every ETH transfer is screened; only matches reach an analyst

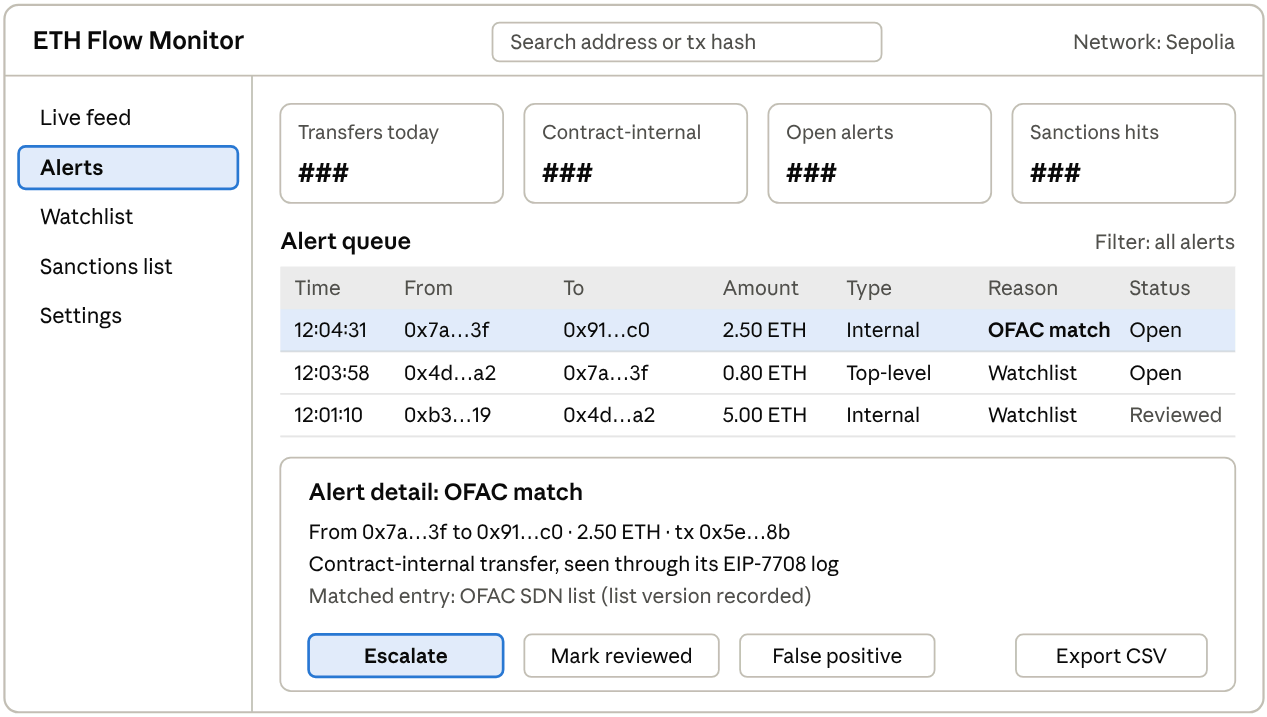


Monitoring flow · from new block to analyst decision

The analyst sets up a watchlist once. After that, every new block is decoded and screened automatically. Transfers with no match show only in the live feed. Matches become an alert that the analyst escalates, marks as reviewed, or marks as a false positive.

Wireframe

Analysts work from one alert queue, with detail one click away



Dashboard wireframe · alerts view, illustrative data

The Alerts view is the main screen. Summary tiles sit on top, the alert queue in the middle, and the selected alert's detail and actions below. Live feed, Watchlist and Sanctions list reuse the same table layout.

Assumptions and constraints

Assumptions

- The EIP-7708 log format stays as it is now. The Glamsterdam meta-EIP (7773) lists 7708 as scheduled for inclusion, and its Last Call period ends November 1, 2026. If the format changes, the decoder changes with it. If 7708 were dropped, the product would fall back to trace-based ingestion and lose its cost advantage.
- The 7708 logs on Sepolia match the spec. Confirmed on October 14, 2026: Glamsterdam has been live on Sepolia since October 6, and the prototype reads its Transfer logs from a free public RPC endpoint, about 25 per block, including contract-internal transfers.

- The OFAC SDN list's published ETH addresses are a good enough sanctions source for an MVP.
- Customers will accept a lightweight monitor alongside, or instead of, a full forensics vendor for real-time ETH alerts.

Constraints

- **Timeline:** 2 months, built by one student.
- **Network:** Sepolia for the demo (Glamsterdam has been live there since October 6, 2026). Mainnet only after Glamsterdam's mainnet fork, which has no confirmed date beyond a Q4 2026 target.
- **History:** no transfer data before the fork, because 7708 logs don't exist there.
- **Budget:** free or student tiers of RPC and hosting providers. The coverage KPI's debug_traceBlock comparison needs trace access, which free tiers often lack (Alchemy's Debug API requires a paid plan). For that test, use a short paid run or your own Sepolia node.
- **Scope:** native ETH on Ethereum L1 only.